MIL-C-29140B(CG)
2 June 1988
SUPERSEDING
MIL-C-29140A(NU)
30 January 1981

MILITARY SPECIFICATION

*CLOTH, PLAIN WEAVE (BASKET VARIATION) POLYESTER AND COTTON (CREASE RESISTANT FINISH)

*This specification is approved for use by the United States Coast Guard and the Department of Transportation and is available for use by all Departments and Agencies.

1. SCOPE

1.1 Scope. This specification covers the requirements for a polyester and cotton blend fabric treated with a crease resistant finish.

Type I - 4.8 oz/yd^2
Type II - Deleted

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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SPECIFICATIONS

*FEDERAL

PPP-P-1134 - Packaging of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

*MILITARY

MIL-T-43548 - Thread, Polyester Core: Cotton-, Rayon-, or Polyester-Covered
MIL-T-43624 - Thread, Polyester, Spun

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

LAWS AND REGULATIONS

Rules and Regulations Under the Textile Fiber Products Identification Act

(Application for copies should be addressed to the Federal Trade Commission, Washington, DC 20580.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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TECHNICAL MANUAL OF THE AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS

Method No. 8 Colorfastness to Crocking: AATCC Crockmeter Method

(Application for copies should be addressed to the AATCC National Headquarters, P.O. Box 12215, Research Triangle Park, NC 27709-9989.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

D 1424 - Tear Resistance of Woven Fabrics by Falling Pendulum (Elmendorf) Apparatus

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard sample. The finished cloth shall match the standard and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3). The standard sample is identified under piece number 79065.

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Materials.

3.3.1 Fiber blend.

3.3.1.1 Polyester fiber. The fiber shall be polyethylene glycol terephthalate, semi-dull, when tested as specified in 4.5.

3.3.1.2 Cotton fiber. The cotton shall be carded and combed, when tested as specified in 4.5.

3.3.2 Yarns. The warp and filling yarns shall be a blend of 65 (±5) percent polyester and the remaining percentage cotton, based on the dry weight of the desized and definished specimens when tested as specified in 4.5. The fibers shall be spun into singles yarn for both the warp and filling (see 4.5).

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* 3.4 Physical requirements. The finished cloth shall conform to the physical requirements shown in Table I when tested as specified in 4.5.

TABLE I. Physical requirements

Characteristics	Minimum	Maximum
Weight oz/yd^2 (g/m^2)	4.8 (162.7)	
Yarns per inch (2.54 cm)		
Warp	92 (92)	
Filling	72 (72)	
Breaking strength lbs (N)		
Warp	110 (489.5)	
Filling	75 (333.8)	
Tearing strength lbs (N)		
Warp	9.5 (42.3)	
Filling	8.5 (37.8)	

3.4.1 Width. The width of the finished cloth shall be as specified (see 6.2) and shall be the minimum acceptable width, inclusive of selvages when fly shuttle looms or shuttleless with tuck-in selvage looms are used. For all other shuttleless looms the width measurement shall be made between the last warp yarn on each side excluding the protruding fringe.

3.4.2 Weave. The weave shall be a modified basket weave as shown in Figure 1, when tested as specified in 4.5.

* 3.5 Color. The color shall be blue shade 3362 and shall match the standard sample.

3.5.1 Sulfur content. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.5.

3.5.2 Matching. The finished cloth shall match the standard sample when viewed under filtered tungsten lamps which approximate artificial daylight having a correlated color temperature of 7500 + 200 K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200 K.

* 3.5.3 Colorfastness. The finished cloth shall show colorfastness to light (40 SFH), laundering (after 3 cycles), perspiration, and crocking equal to or better than the standard sample. As a limit of acceptability or when no standard sample is available, the finished cloth shall show a minimum of "good" fastness to light (40 SFH), laundering (after 3 cycles), and perspiration, and shall show a dry crocking rating not lower than 4 and a wet crocking not lower than 3 when tested as specified in 4.5.

3.6 Nonfibrous material. The total nonfibrous material content of resin finished cloth shall not exceed 6.0 percent when tested as specified in 4.5.

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3.7 pH. The pH value of the water extract of the finished cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.5.

3.8 Dimensional stability. The finished cloth shall not shrink or elongate more than 2.0 percent in either the warp or filling direction when tested as specified in 4.5. The preshrinking process used shall not be identified by name or by trademark either on the cloth, ticket, or package.

3.9 Seam efficiency. The cloth shall have a seam efficiency of not less than 85 percent when tested as specified in 4.5.

3.10 Finish. The cloth shall be desized, scoured, dyed, heat set, treated with a resin and cured to provide a crease-resistant finish and hand equal in characteristic to that of the standard sample, when tested as specified in 4.5.

3.11 Length and put-up. Unless otherwise specified (see 6.2), the finished cloth shall be furnished in continuous lengths, each not less than 40 yards (36.4 m). Each length shall be put-up on full width rolls as specified in PPP-P-1134 with the face inward.

3.12 Marking. The face of the cloth shall be identified by stamping on that side with the word "FACE" at each end of each piece.

3.13 Fiber identification. Each piece shall be labeled, or ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.14 Workmanship. The finished cloth shall conform to the quality established by this document. The occurrence of defects shall not exceed the applicable point level specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required (see 3.2 and 6.2), the first article shall be visually examined for appearance, color and finish, and shall be tested for physical and chemical properties in accordance with methods specified in 4.5.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.4.

4.4.2.1 Yard-by-yard examination. The required yardage of each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards (91.4m), the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards (91.4m), only 100 yards (91.4m) shall be examined. All defects as defined in Section 1 of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) (0.91m) shall be scored and assigned demerit points as listed in 4.4.2.1.1, except that only those slubs and knots which exceed the limits shown on Sears Fabric Defect Scale (see 6.4), "C" or "4" for slubs and "B" for knots, shall be scored. No linear yard (increments of 1 yard (0.9m) on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards ($83.6m^2$) of the total yardage examined exceeds 25 points. The lot shall be unacceptable if the points per 100 square yards ($83.6m^2$) of two or more individual rolls exceed 38 points. If one roll exceeds the point level, a second sample of 20 rolls shall be examined only for individual roll quality examination. If one or more rolls exceed the value, the lot shall be rejected. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards}$$

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* 4.4.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects up to and including 3 inches in any dimension	- one point
For defects exceeding 3 inches but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

The following defects, when present, shall be scored four points for each yard (0.9m) in which they occur.

Baggy, ridgy, or wavy cloth
Width less than specified
Characteristics of finish, hand and shade not equal to the standard sample
Poor dye penetration, mottled, streaky, or cloudy
Excessive neppiness

4.4.2.2 Examination for length.

4.4.2.2.1 Individual rolls. During the yard by yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified, or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.4.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual length of rolls in the sample is less than the total of the lengths marked on the ticket. The rolls examined shall be those selected for the examination of individual rolls.

* 4.4.2.3 Examination for shade variation. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation. Any roll in the sample exhibiting uneven shade, shaded variation side to side, side to center or end to end shall be cause for rejection of the entire lot represented by the sample.

4.4.2.4 Examination for face identification, identification of preshrinkage process and non-conformance with the Textile Fiber Products Identification Act. During the yard by yard examination, each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample have face identification missing from either or both ends, two or more rolls in the sample contain identification of the preshrinkage process by name or trademark on the cloth or ticket or two or more rolls in the sample are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.4.3 Examination of packaging requirements. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that preservation, packaging, packing, and marking comply with section 5 requirements.

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* 4.5 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in Table II shall be followed. The physical and chemical values specified in Section 3, except where otherwise specified, apply to the average of the determinations made on the sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing final results. The sample unit shall be 4 1/2 yards (4.1m), full width of the finished cloth. The lot size shall be expressed in units of 1 yard. The sample size shall be as follows and the lot shall be unacceptable if one or more sample units fail to meet any requirements specified.

Lot size yards	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II. - Test Methods

Characteristic	Requirement Paragraph	Test Method
Fiber identification	3.3.1	1/
Fiber content:		
Polyester (percent)	3.3.2	2100, 4.5.1
Cotton (percent)	3.3.2	2100, 4.5.1
Yarns (singles), warp and filling	3.3.2	Visual 2/
Weight	3.4	5041
Yarns per inch, warp and filling	3.4	5050
Breaking strength, warp and filling	3.4	5100
Tear strength, warp and filling	3.4	ASTM D-1424
Weave	3.4.2	Visual 2/
Sulfur content	3.5.1	2020
Colorfastness:		
Light	3.5.3	5660 3/
Laundering (after 3 cycles)	3.5.3	5610 4/ 5/
Perspiration	3.5.3	5680
Crocking	3.5.3	AATCC-TM 8
Nonfibrous materials	3.6	2610
pH	3.7	2811
Dimensional stability	3.8	5556 6/
Seam efficiency	3.9	5110 7/
Finish	3.10	1/

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirements.
2/ One determination per sample unit and the results reported as "pass" or "fail".

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3/ The calibration of the fadeometer and the definition of AATCC Fading Units shall be according to AATCC Test Method 16-1982. Using the formula below, 16 to 22 AFU's are acceptable as 20 SFH's.

$$\text{AATCC Fading Units} = \frac{(\Delta E^*)\ (20)}{1.7}$$

Where (ΔE^*) is the color difference in CIELAB* units of color difference.
4/ The specimens must be dried after each of the 3 laundering cycles.
5/ Only the stain on the cotton and polyester fibers of the color transfer cloth shall be evaluated.
6/ Cotton laundering procedure shall be followed.
7/ The needle shall measure 0.040 inch (±0.001) across the blade at the eye. The thread shall be either polyester/cotton in accordance with MIL-T-43548, ticket no. 50, 2 ply for the needle and ticket no. 70, 2 ply for the looper or spun polyester in accordance with MIL-T-43624, class 1 or 2, ticket no. 40, 3 ply for the needle and ticket no. 70, 2 ply for the looper.

4.5.1 Polyester and cotton fiber content. The general procedures of method 2100 shall be followed with the fiber content percentages calculated as follows:

$$\text{Percent polyester} = \frac{\text{Weight of dry residual fiber} \times 100}{\text{Weight of dry desized specimen}}$$

Percent cotton = 100 - percent polyester

4.5.1.1 Report. Two specimens shall be tested from each sample unit and the average percent polyester and the average percent cotton of the two specimens shall be reported to the nearest 1.0 percent.

5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1134.

6. NOTES

* 6.1 Intended use. The cloth covered by this specification is intended for use in the blue utility shirt worn by Coast Guard personnel.

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* 6.2 Ordering data. Procurement documents should specify the following:
a. Title, number and date of this document.
b. When a first article is required (see 3.2 and 4.3). The item will be tested and should be a first article sample. The contracting officer should include specific instruction in acquisition documents regarding arrangements for examinations, quantity, testing, and approval of the first article.
c. Width required (see 3.4.1).
d. Length required if other than specified (see 3.11).
e. Levels of put-up and preservation, and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids (see 3.1).

6.4 Sears Fabric Defect Scale. Sears Fabric Defect Scales are available from Sears, Roebuck and Company, Department 817, (Attn: 23-29), Sears Tower, Chicago, IL 60684 (see 4.4.2.1).

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluadte the requirements of theis document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* 6.7 Subject term (key word) listing.
Cloth, basket weave
Cloth, crease-resistant finish
Cloth, polyester and cotton
Finish, crease-resistant

Custodian:
Navy - NU

Review Activity:
DLA - CT
Navy - CG

Preparing Activity:
Navy - NU

Project No.
8305-N234

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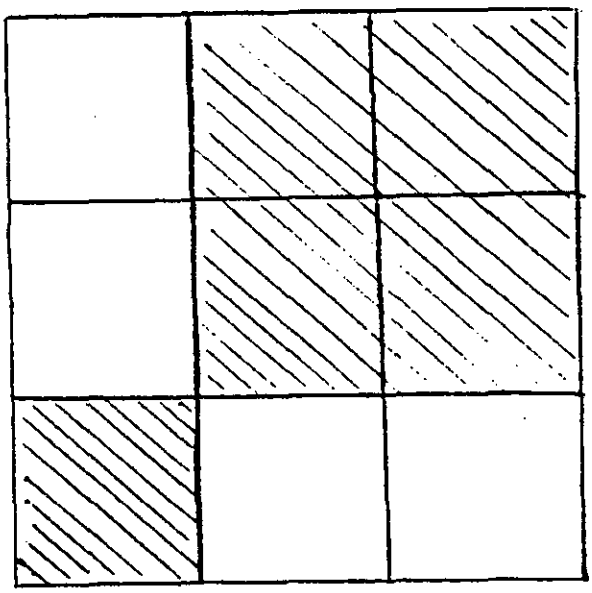


FIGURE 1

Weave Diagram

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